

Jan 23, 2025 | 📅 RV Perf Analysis SIG

Attendees: Beeman Strong tech.meetings@riscv.org Snehasish Kumar

Notes

- **Attendees:** Beeman, Bruce, Chun, MattT, Snehasish
- Slides/video [here](#)
- Meeting changes
 - Moving to using LFX for scheduling, Confluence for storage
 - Voting rights, 1 vote per org/company
- TG updates
 - Perf Events TG hopes to start internal review (IR) this quarter
 - Perf Event Sampling TG still in definition, IR in Q2
 - Self-hosted Trace TG in early stages, hoping for IR in Q2
 - All aiming for ratification this year
- Reviewing gaps and priorities
 - See details in [gap analysis gsheet](#)
 - Last prioritization order (from 2023), and status
 - CTR - resolved, extension ratified
 - Counter Access (S/VS) - deleg to S ratified, no deleg to VS yet
 - Standard Events - Perf Events TG in place
 - Hardware Sampling - Perf Sampling TG in place
 - Standard SoC PMU
 - Standard Counters
 - Precise Sampling - Perf Sampling TG in place
 - Event Repo - Perf Events TG in place
 - Event Modifiers
 - Counter Freeze - discussed in sampling TG, no demand for this
 - Analysis Methodology - Perf Events TG in place
 - Proposed new order
 - Precise Sampling
 - Standard Events
 - Hardware Sampling
 - Standard SoC PMU
 - Counter Access (VS)
 - Analysis Methodology
 - Event Repo
 - Event Modifiers
 - Standard Counters
 - Counter Freeze
 - SiFive has HW-resettable counters
 - Discussed some new capabilities in industry (all from Intel)
 - Auto Counter Reload

- Overflow of one counter can cause HW reload of itself and/or other counters
 - Enables rate-based sampling
- ii. LBR Event Logging
 - 4 2-bit saturating counters per LBR entry, leverages event selector for lower programmable counters
 - Provides some cause info to go with cycle time
- iii. PT Trigger Tracing
 - Akin to Sdtrig integration with trace, but also perf counters can cause trace start/stop/notify
- iv. Counter read-and-clear
 - Specific for PERF_METRICS reg, RV doesn't have anything like this
- Event trace kind of like LBR logging
- Discussed snapshotting counter in Event Trace every so often
- Should consider usability as much as new features
 - i. Challenges around virt
 - Pass-through PMU has challenges
 - Might want to partition counters (guest vs host)
 - VS-mode deleg would support that
 - PEBS buffer is challenging too
 - Self-hosted trace TG is discussing addressing for memory buffer, considering virt and nested virt. Sampling TG will leverage the same buffer
 - RV has benefit of defining capabilities with virt in mind
 - Want to be able to monitor per QoS CoS
 - i. Believe MCID allows this, for measurements like memory BW
- Out of time. Will start email threads to consider additions to gap list, and cover SW gaps/priorities (not reached today).

Action items

- ☐ Beeman Strong - Jul 28, 2022 - Reach out about proprietary performance analysis tools